

PROJECT CONTROLS

The CPI that lies

A cost performance index can look excellent and still be wrong. Here is the arithmetic.

Month-end on a control account

Earned value 2,200,000.

Invoiced cost so far 1,850,000.

A subcontractor has done a further 240,000 of work. It has not been invoiced yet.

Skip the accrual

$$\begin{aligned}\text{CPI} &= 2,200,000 \div 1,850,000 \\ &= 1.19\end{aligned}$$

Comfortably under budget. The report writes itself.

Now book the accrual

$$\begin{aligned} AC &= 1,850,000 + 240,000 \\ &= 2,090,000 \end{aligned}$$

$$\begin{aligned} CPI &= 2,200,000 \div 2,090,000 \\ &= 1.05 \end{aligned}$$

0.14

The index moved fourteen points on one missing accrual. Nothing about the work changed.

Why it matters

Every forecast built on CPI inherits the error. An estimate at completion computed on 1.19 is not slightly optimistic; it is answering a different question from the one the board asked.

The professional act

Actual cost is not what has been invoiced. It is what has been *incurred*: invoices, plus commitments delivered and not yet billed, plus accruals, at a stated cut-off.

Cut-off is a decision. Record it.

PCI STANDARDS

Accrual completeness

A performance index reported without a stated cut-off and a complete accrual position is not a measurement. It is an impression.

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